

Data Science Program, George Washington University

Sample Exam 1

DATS 6103 (Introduction to Data Mining)

Date: **October 27, 2026**

Instructor: Prof. Sushovan Majhi

Name: _____

Please read the following instructions carefully.

- This is a **sample paper** for practice only, at the real length and the real difficulty, covering everything through Oct 20 but **not** regression as prediction.
- This paper prints **30 problems**. **Answer any 25**. Each is worth **6 points**, so the exam is marked out of **150**. If you answer more than 25, **your best 25 count**.
- You have **2 hours 30 minutes**, the full session.
- Most problems are multiple choice and **exactly one option is correct**. Mark it clearly (✓). The rest say what they want.
- The exam is **open-book** and **open-note**, but **printed material only**: the chapters covered, the labs whose solutions have been released, the homework covered with its solutions, and your own notes.
- **Laptops, tablets, phones** and **watches** must be **switched off and put away**. No AI assistance and no communication of any kind.
- **Nothing has to be executed**. Code you write is graded on whether it would do the right thing, not on a missing colon; a shape, a dtype or the name of an error earns marks.
- All notation has its usual meaning. You may use standard results directly by stating them appropriately. **Read the instructions for individual questions carefully** before answering.

Question	Points	Score
1	6	
2	6	
3	6	
4	6	
5	6	
6	6	
7	6	
8	6	
9	6	
10	6	
11	6	
12	6	
13	6	
14	6	
15	6	
16	6	
17	6	
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19	6	
20	6	
21	6	
22	6	
23	6	
24	6	
25	6	
26	6	
27	6	
28	6	
29	6	
30	6	
Total:	180	

Part A Python

1. (6 points) `a = [1, 2, 3]`

```
2 b = a
3 b.append(4)
4 print(len(a))
```

prints

- ☐ 3
- ☐ 4
- ☐ a `TypeError`
- ☐ nothing; `append` returns `None`

2. (6 points) `def tag(item, seen=[]):`

```
2     seen.append(item)
3     return seen
4 tag("x")
5 print(tag("y"))
```

prints

- ☐ `['y']`
- ☐ `['x', 'y']`
- ☐ `['x']`
- ☐ a `NameError`

3. (6 points) `x = [3, 1, 2]` and `y = x.sort()`. What is `y`?

- ☐ `[1, 2, 3]`
- ☐ `None`
- ☐ `[3, 1, 2]`
- ☐ a `TypeError`

4. (6 points) `d = {"b": 2, "a": 1}`. What does `list(d)` give?

- ☐ `['b', 'a']` — the keys in insertion order
- ☐ `['a', 'b']` — the keys, sorted
- ☐ `[2, 1]` — the values
- ☐ an arbitrary order that may differ between runs

5. (6 points) **Short answer.** This is meant to return a cleaned copy and leave its argument alone. Say what goes wrong, and write a corrected body.

```
1 def drop_negatives(values):
2     for i, v in enumerate(values):
3         if v < 0:
4             values[i] = 0
5     return values
```

Part B NumPy

6. (6 points) The shape of `np.zeros((4, 1)) + np.zeros((3,))` is
- ☐ (4, 3)
 - ☐ (4, 1)
 - ☐ (3, 4)
 - ☐ it raises a `ValueError`
7. (6 points) T has shape (30, 12, 4) and B has shape (12,). `T + B`
- ☐ broadcasts to (30, 12, 4)
 - ☐ raises, because 12 is compared against the last axis, which is 4
 - ☐ broadcasts to (30, 12, 12)
 - ☐ broadcasts, but silently truncates
8. (6 points) A is a contiguous 2-D array. Which returns a **copy** rather than a view?
- ☐ `A[1:3]`
 - ☐ `A.T`
 - ☐ `A[[0, 2]]`
 - ☐ `A.reshape(-1)`
9. (6 points) T is (30, 12, 4): days, sensors, channels. `T.mean(axis=1)` returns
- ☐ shape (30, 4) — one value per day per channel
 - ☐ shape (12, 4) — one value per sensor per channel
 - ☐ shape (30, 12) — one value per day per sensor
 - ☐ a single number
10. (6 points) `T.mean(axis=(0, 1), keepdims=True)` returns shape
- ☐ (4,)
 - ☐ (1, 1, 4)

- ☐ (30, 12, 1)
 - ☐ (1, 4)
11. (6 points) `A = np.arange(6).reshape(2, 3)`. Then `A.T.ravel()` is
- ☐ `[0 1 2 3 4 5]`
 - ☐ `[0 3 1 4 2 5]`
 - ☐ `[5 4 3 2 1 0]`
 - ☐ `[0 3 1 4 2 5]` as a view on `A`
12. (6 points) **Short answer.** Of `T[0]`, `T.reshape(-1, 4)`, `T[T[..., 0] > 25]` and `T.T`, say which share memory with `T`, and in one sentence what the ones that share have in common.

Part C Linear algebra in NumPy

13. (6 points) `A` is (15, 4). Then `A @ A`
- ☐ has shape (15, 15)
 - ☐ has shape (4, 4)
 - ☐ raises, because the inner dimensions 4 and 15 disagree
 - ☐ has shape (15, 4)
14. (6 points) With the same `A`, the shapes of `A.T @ A` and `A @ A.T` are
- ☐ (15, 15) and (4, 4)
 - ☐ (4, 4) and (15, 15)
 - ☐ both (4, 4)
 - ☐ both (15, 15)
15. (6 points) If $\text{cond}(A) = 10^4$, then $\text{cond}(A^T A)$ is about 10^8 , which matters because
- ☐ the matrix becomes singular
 - ☐ you lose roughly $\log_{10}$ of the condition number in digits, so the normal equations spend about eight of double precision's sixteen before any arithmetic
 - ☐ the product is no longer symmetric

☐ solve will refuse to run

16. (6 points) X is $(100, 5)$ and $U, S, Vt = \text{np.linalg.svd}(X, \text{full_matrices}=\text{False})$. The shape of S is

- ☐ $(5, 5)$
- ☐ $(5,)$
- ☐ $(100, 5)$
- ☐ $(100,)$

Part D Pandas

17. (6 points)

```
q1 = pd.Series([100, 120, 90], index=["shaw", "petworth", "
    navy yard"])
q2 = pd.Series([110, 95, 130], index=["petworth", "navy yard", "
    brookland"])
total = q1 + q2
```

`total` has

- ☐ 3 entries, none missing
- ☐ 3 entries, one missing
- ☐ 4 entries, two missing
- ☐ 6 entries, none missing

18. (6 points) $s = \text{pd.Series}([10, 20, 30, 40], \text{index}=[2, 3, 4, 1])$. Then $s.\text{loc}[2]$ and $s.\text{iloc}[2]$ are

- ☐ 10 and 30
- ☐ 30 and 10
- ☐ both 10
- ☐ both 30

19. (6 points) $s = \text{pd.Series}([1, 2, 3], \text{index}=["a", "b", "c"])$. How many entries does $s["b": "c"]$ return?

- ☐ 1
- ☐ 2 — label slicing includes the right endpoint
- ☐ 3
- ☐ it raises, because slicing needs integers

20. (6 points) df has shape $(5, 3)$. $df.\text{mean}(\text{axis}=1)$ returns

- ☐ 3 numbers, one per column
- ☐ 5 numbers, one per row

- ☐ a (5, 3) frame
 - ☐ a single number
21. (6 points) A column holds [12.0, 8.0, NaN]. Its `.mean()` is
- ☐ NaN, because one value is missing
 - ☐ 10.0 — the gap is skipped and the divisor is 2
 - ☐ 6.67 — the gap counts as zero
 - ☐ 10.0, and `skipna=False` would give the same
22. (6 points) `orders` has customer b once; `lookup` has customer b twice. Merging on `cust` produces, for b,
- ☐ 1 row, keeping the first match
 - ☐ 2 rows, and any total over that column is now double-counted
 - ☐ 0 rows, because the key is ambiguous
 - ☐ an error
23. (6 points) `orders["size"].map({"S": 1, "M": 2, "L": 3})` on a column containing "XL" gives
- ☐ a `KeyError`
 - ☐ a float64 column with NaN where XL was
 - ☐ an int64 column with 0 where XL was
 - ☐ the original strings, unmapped
24. (6 points) `tickets` logs 3 and 5 hours for core in week 1. `pivot_table(index="team", columns="w", gives, for core in w1,`
- ☐ 8, the sum
 - ☐ 4, the mean, because that is the default `aggfunc`
 - ☐ 3, the first value
 - ☐ an error, because the pair is duplicated
25. (6 points) **Short answer.** Three orders totalling \$97 are merged against a lookup table in which one customer appears twice. The merged frame has four rows and its total sums to \$152. Explain what happened, and name the argument that would have raised instead.

Part E Exploratory analysis

26. (6 points) A histogram of 3,000 rents drawn with five bins looks like a single mound. The safest reading is
- ☐ the rents are unimodal
 - ☐ the bin width is a choice, and five bins cannot show structure narrower than a bin
 - ☐ the sample is too small to judge
 - ☐ a histogram is the wrong plot for rents
27. (6 points) Mean rent \$2,180, median rent \$1,740. This says the distribution is
- ☐ right-skewed, with a minority of high rents pulling the mean up
 - ☐ left-skewed
 - ☐ symmetric, with the difference due to rounding
 - ☐ bimodal
28. (6 points) **Short answer.** Give two distinct reasons the five-bin conclusion above might not survive a different chart, and name one summary statistic you would ask for alongside it.

Part F Statistical learning

29. (6 points) A model scores 0.02 mean squared error on its training data and 0.35 on data it has not seen. This is
- ☐ overfitting, and the 0.02 is worth nothing as a performance claim
 - ☐ underfitting, and the model needs more flexibility
 - ☐ a bug, since test error cannot exceed training error
 - ☐ acceptable, since the average of the two is reasonable
30. (6 points) **Short answer.** In one or two sentences, say what making a model more flexible does to bias and to variance, and why the best model is not the most flexible one.

SAMPLE

*Thirty problems printed; your best twenty-five count.
Code is graded on whether it would do the right thing, not on a missing colon.*